

LECTURE 1

Mathematical preliminaries

Foundations of Artificial Intelligence

SEMTM0051 · 2026/27

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Today's lecture

- Matrices
- Differential calculus
- Finding minima using differential calculus

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- **Matrices**
- Differential calculus
- Finding minima using differential calculus

Scalars, vectors and matrices

A matrix is a rectangular array of numbers, arranged in m rows and n columns.

$$\mathbf{A} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

A_{ij} is the scalar entry in row i and column j .

$$1 \leq i \leq m, \quad 1 \leq j \leq n$$

Vectors are matrices with exactly one row or one column.

$$\mathbf{r} = (r_1 \quad \cdots \quad r_d) \in \mathbb{R}^{1 \times d}$$

$$\mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_d \end{pmatrix} \in \mathbb{R}^{d \times 1}$$

Adding matrices

Matrices with the same shape can be added:

Add each entry of the first matrix to the entry in the same position of the second

In general, add matching entries:

$$(\mathbf{A} + \mathbf{B})_{ij} = A_{ij} + B_{ij}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

The sum has the same shape as **A** and **B**. Subtraction works entry by entry too.

Multiplying a matrix by a scalar

Multiplying a matrix by a number means multiplying every entry by that number.

The shape stays the same.

For any scalar λ :

$$(\lambda \mathbf{A})_{ij} = \lambda A_{ij}$$

$$3 \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 & 3 \cdot (-2) \\ 3 \cdot 0 & 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 3 & -6 \\ 0 & 12 \end{pmatrix}$$

The same rules apply to row and column vectors, because they are matrices too.

Transposes

Transposing turns each row into a column.
The entries keep their values and change position.

$$(\mathbf{B}^T)_{ij} = B_{ji}$$

In general, a matrix with shape $m \times n$ becomes one with shape $n \times m$.

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad \mathbf{B}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

The same operation turns a row vector into a column vector.

$$(1 \quad 2)^T = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Matrix multiplication

Each output entry combines one row of the first matrix with one column of the second. Hence, their lengths must agree.

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{B}}_{n \times p} = \underbrace{\mathbf{C}}_{m \times p}$$

For entry C_{ik} , multiply the paired entries and add their products.

$$C_{ik} = \sum_{j=1}^n A_{ij} B_{jk}$$

The index j runs through the pairs. The indices i and k identify the output.

Matrix multiplication: following the arrows

For each output entry, move along a row of **A** and down a column of **B**.
Multiply the entries you meet together.

$$C_{11} : \begin{pmatrix} \rightarrow & \rightarrow \end{pmatrix} \begin{pmatrix} \downarrow \\ \downarrow \end{pmatrix}$$

$$C_{12} : \begin{pmatrix} \rightarrow & \rightarrow \end{pmatrix} \begin{pmatrix} \downarrow \\ \downarrow \end{pmatrix}$$

$$C_{21} : \begin{pmatrix} \rightarrow & \rightarrow \end{pmatrix} \begin{pmatrix} \downarrow \\ \downarrow \end{pmatrix}$$

$$C_{22} : \begin{pmatrix} \rightarrow & \rightarrow \end{pmatrix} \begin{pmatrix} \downarrow \\ \downarrow \end{pmatrix}$$

The output position selects the row and column.
The arrows show how to pair their entries.

Computing C_{11} : row 1, column 1

C_{11} uses row 1 of A and column 1 of B. The two colours will identify the two pairs of factors.

$$\begin{array}{c} \text{A} \\ \left[\begin{array}{cc} 5 & 6 \\ 7 & 8 \end{array} \right] \end{array} \times \begin{array}{c} \text{B} \\ \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \end{array} = \begin{array}{c} \text{C} \\ \left[\begin{array}{c} \boxed{23} \end{array} \right] \end{array}$$

$$C_{11} = 5 \times 1 + 6 \times 3 = 23$$

$5 + 18 = 23$: the sum fills the selected entry of C.

Computing C_{12} : row 1, column 2

C_{12} uses row 1 of A and column 2 of B. The two colours will identify the two pairs of factors.

$$\begin{array}{c} \text{A} \\ \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \end{array} \times \begin{array}{c} \text{B} \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \end{array} = \begin{array}{c} \text{C} \\ \begin{bmatrix} 23 & \boxed{34} \end{bmatrix} \end{array}$$

$$C_{12} = 5 \times 2 + 6 \times 4 = 34$$

$10 + 24 = 34$: the sum fills the selected entry of C.

Computing C_{21} : row 2, column 1

C_{21} uses row 2 of A and column 1 of B. The two colours will identify the two pairs of factors.

$$\begin{array}{c} \text{A} \\ \left[\begin{array}{cc} 5 & 6 \\ 7 & 8 \end{array} \right] \end{array} \times \begin{array}{c} \text{B} \\ \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \end{array} = \begin{array}{c} \text{C} \\ \left[\begin{array}{cc} 23 & 34 \\ \boxed{31} & \end{array} \right] \end{array}$$

$$C_{21} = 7 \times 1 + 8 \times 3 = 31$$

$7 + 24 = 31$: the sum fills the selected entry of C.

Computing C_{22} : row 2, column 2

C_{22} uses row 2 of A and column 2 of B. The two colours will identify the two pairs of factors.

$$\begin{array}{c} \text{A} \\ \left[\begin{array}{cc} 5 & 6 \\ 7 & 8 \end{array} \right] \end{array} \times \begin{array}{c} \text{B} \\ \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \end{array} = \begin{array}{c} \text{C} \\ \left[\begin{array}{cc} 23 & 34 \\ 31 & \boxed{46} \end{array} \right] \end{array}$$

$$C_{22} = 7 \times 2 + 8 \times 4 = 46$$

$14 + 32 = 46$: the sum fills the selected entry of C.

Matrix multiplication is not commutative

Reversing the order of the factors can change whether the product is defined, and it can change its value.

$$\begin{array}{c} \text{A} \\ \left[\begin{array}{cc} 5 & 6 \\ 7 & 8 \end{array} \right] \end{array} \times \begin{array}{c} \text{B} \\ \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \end{array} = \begin{array}{c} \text{C} \\ \left[\begin{array}{cc} 23 & 34 \\ 31 & 46 \end{array} \right] \end{array}$$

Warning: AB can be defined even when BA is not.

Even when both have the same shape, in general $AB \neq BA$.

Exercise: compute BA for the matrices above and compare it with AB.

Properties of matrix multiplication

For compatible dimensions, changing the brackets does not change the product.

Matrix multiplication is **associative**.

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

If you have never checked this, prove it as an exercise!

Transposing a product reverses the order of its factors.

$$(\mathbf{AB})^{\top} = \mathbf{B}^{\top} \mathbf{A}^{\top}$$

The Euclidean norm

The **Euclidean norm** is the square root of the sum of the squared entries.
Its square can also be written as a matrix product.

For a column vector $\mathbf{c} \in \mathbb{R}^{d \times 1}$:

$$\|\mathbf{c}\|_2^2 = \sum_{j=1}^d c_j^2 = \mathbf{c}^\top \mathbf{c}$$

For a row vector $\mathbf{r} \in \mathbb{R}^{1 \times d}$:

$$\|\mathbf{r}\|_2^2 = \sum_{i=1}^d r_i^2 = \mathbf{r} \mathbf{r}^\top$$

Both products have shape 1×1 , which we identify with a scalar.

The identity matrix

The **identity** matrix $\mathbf{I}_n$ is the $n \times n$ matrix with ones on the diagonal and zeros everywhere else.

$$\mathbf{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix} \in \mathbb{R}^{n \times n}$$

Multiplication leaves $\mathbf{A} \in \mathbb{R}^{m \times n}$ unchanged when the dimensions match:

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} = \mathbf{A} \mathbf{I}_n$$

If you have never checked this, prove it as an exercise!

Invertible matrices

A **square** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $\mathbf{B} \in \mathbb{R}^{n \times n}$ s.t.

$$\mathbf{AB} = \mathbf{I}_n = \mathbf{BA}$$

One can prove that this matrix $\mathbf{B}$ is **unique**.

We call it the **inverse** of $\mathbf{A}$ and denote it by $\mathbf{A}^{-1}$.

Thus

$$\mathbf{AA}^{-1} = \mathbf{I}_n = \mathbf{A}^{-1}\mathbf{A}$$

A square matrix with no inverse is called **singular**.

An inverse in two dimensions

For a 2×2 matrix, the inverse exists exactly when $ad - bc \neq 0$. In that case,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

For example, here $ad - bc = 4 - 6 = -2$:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Check that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_2 = \mathbf{A}^{-1}\mathbf{A}$.

Solving a linear system with an inverse

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible and $\mathbf{b} \in \mathbb{R}^{n \times 1}$.

We want to solve the linear system $\mathbf{Ax} = \mathbf{b}$ for a column vector $\mathbf{x}$.

Any solution must satisfy:

$$\mathbf{x} = \mathbf{I}_n \mathbf{x} = (\mathbf{A}^{-1} \mathbf{A}) \mathbf{x} = \mathbf{A}^{-1} (\mathbf{Ax}) = \mathbf{A}^{-1} \mathbf{b}$$

This candidate also satisfies the equation:

$$\mathbf{A}(\mathbf{A}^{-1} \mathbf{b}) = (\mathbf{AA}^{-1}) \mathbf{b} = \mathbf{I}_n \mathbf{b} = \mathbf{b}$$

Therefore the solution exists and is unique: $\mathbf{x} = \mathbf{A}^{-1} \mathbf{b}$.

Indicator functions and the Kronecker delta

An indicator turns a condition into a number:
one when the condition holds, and zero otherwise.

$$\mathbf{1}\{A\} := \begin{cases} 1, & \text{if } A \text{ is true,} \\ 0, & \text{if } A \text{ is false.} \end{cases}$$

The Kronecker delta is the indicator that two indices agree:

$$\delta_{ij} = \mathbf{1}\{i = j\}, \quad (\mathbf{I}_d)_{ij} = \delta_{ij}$$

In a sum, it keeps only the term with the matching index:

$$\sum_{j=1}^d \delta_{ij} a_j = 0a_1 + \cdots + 0a_{i-1} + 1a_i + 0a_{i+1} + \cdots + 0a_d = a_i$$

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- Finding minima using differential calculus

Functions and function values

A function assigns exactly one output to each input in its domain. The graph shows how inputs and outputs are paired.



$$f : \mathbb{R} \longrightarrow \mathbb{R}$$

$$f(x) = x^2$$

At input $x = 3$, the output is

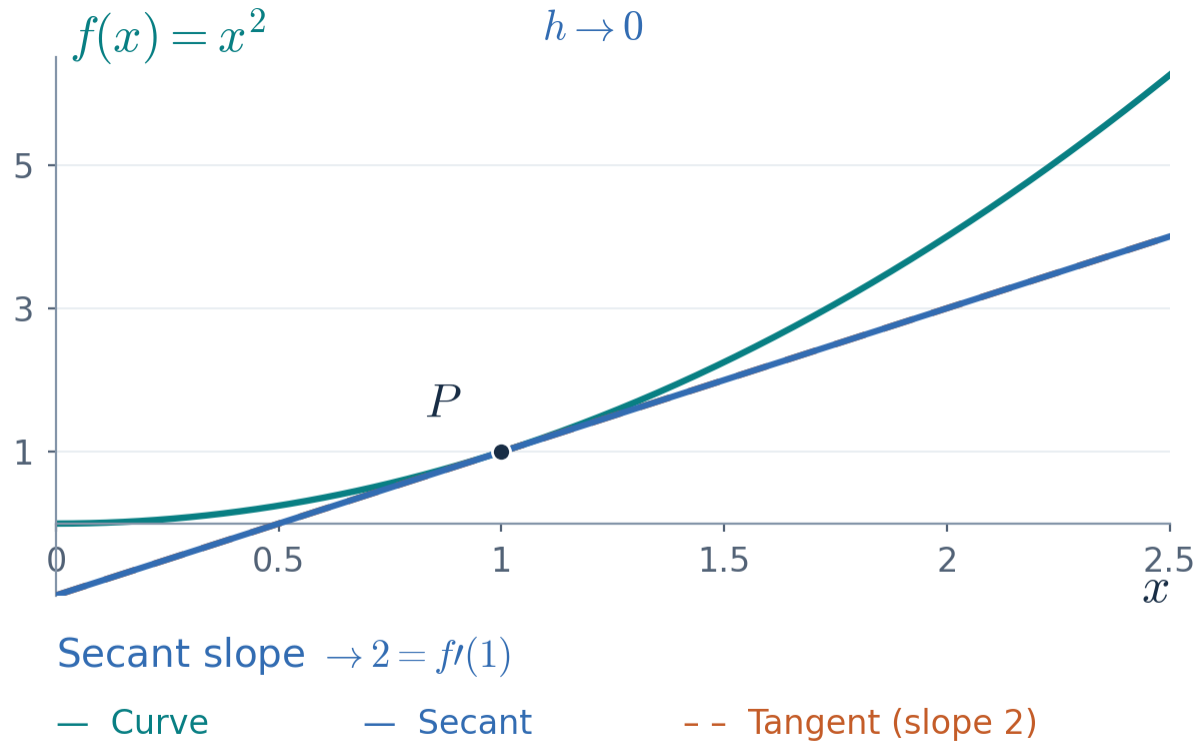
$$f(3) = 3^2 = 9$$

The highlighted point is (3,9).

The graph consists of the points $(x, f(x))$.

A derivative measures change

A derivative is the limiting slope as the two points of a secant come together.



The secant uses two points. As the points come together, it approaches the tangent.

$$\begin{aligned} f'(x) &= \frac{df}{dx}(x) \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \end{aligned}$$

$$f'(1) = 2$$

The derivative exists when this limit is a finite real number.

Standard derivatives

Constants, powers, exponentials and logarithms give the basic rules.

$$\frac{d}{dx}c = 0$$

$$\frac{d}{dx}e^x = e^x$$

$$\frac{d}{dx}x^p = px^{p-1}$$

$$p \neq 0, \quad x > 0$$

$$\frac{d}{dx} \log x = \frac{1}{x} \quad (x > 0)$$

If $p \in \mathbb{Z}$, the power rule also holds for $x \neq 0$.

Linearity and the product rule

For constants a and b , differentiate each term and keep its coefficient.

$$[a u(x) + b v(x)]' = a u'(x) + b v'(x)$$

$$\frac{d}{dx} (3x^4 + 2x^{-1/2} + x^{-2}) = 12x^3 - x^{-3/2} - 2x^{-3} \quad (x > 0)$$

For a product, differentiate each factor in turn and keep the other fixed.

$$[u(x)v(x)]' = u'(x)v(x) + u(x)v'(x)$$

$$\frac{d}{dx} [x^2(x + 1)] = 2x(x + 1) + x^2 \cdot 1$$

The chain rule: two notations

Let $F(s) = v(u(s))$, i.e. F is the composition of v and u , $F = v \circ u$

$$s \longmapsto u(s) \longmapsto v(u(s))$$

Prime notation

$$F'(s) = v'(u(s)) u'(s)$$

Leibniz notation

$$\frac{dF}{ds}(s) = \left. \frac{dv}{du} \right|_{u=u(s)} \frac{du}{ds}(s)$$

Abbreviated form
(abuse of notation)

$$\frac{dv}{ds} = \frac{dv}{du} \frac{du}{ds}$$

Here dv/du is evaluated at $u = u(s)$,
while dv/ds denotes the derivative of the composition F .

A complete chain-rule example

Differentiate the inner and outer functions, then substitute and multiply.

$$F(s) = (2s + 5)^3$$

$$u(s) = 2s + 5, \quad v(u) = u^3$$

$$u'(s) = 2, \quad v'(u) = 3u^2$$

$$F'(s) = v'(u(s))u'(s) = 3(2s + 5)^2 \cdot 2$$

$$\frac{dF}{ds}(s) = \left. \frac{dv}{du} \right|_{u=2s+5} \frac{du}{ds}(s) = 3(2s + 5)^2 \cdot 2 = 6(2s + 5)^2$$

Exercise: differentiate $H(s) = \log(1 + e^s)$ **using both notations.**

Partial derivatives: one variable at a time

Fix all inputs except one, then take an ordinary derivative.
The resulting partial derivative can still depend on every input.

$$\partial_1 F(a, b) = \frac{\partial F}{\partial a}(a, b) = \lim_{h \rightarrow 0} \frac{F(a + h, b) - F(a, b)}{h}$$

$$\partial_2 F(a, b) = \frac{\partial F}{\partial b}(a, b) = \lim_{h \rightarrow 0} \frac{F(a, b + h) - F(a, b)}{h}$$

Each partial derivative exists when its limit is a finite real number.

Example: $F(a, b) = a^2 + 3ab + b^2$

Vary a and keep b fixed.

$$\partial_1 F(a, b) = \frac{\partial F}{\partial a}(a, b) = 2a + 3b$$

$$\partial_1 F(1, 2) = 8$$

Vary b and keep a fixed.

$$\partial_2 F(a, b) = \frac{\partial F}{\partial b}(a, b) = 3a + 2b$$

$$\partial_2 F(1, 2) = 7$$

Differentiability

A function $Q: \mathbb{R}^d \rightarrow \mathbb{R}$ is **differentiable** at $\mathbf{w}$ if there exists a column vector $\mathbf{g} \in \mathbb{R}^d$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|Q(\mathbf{w} + \mathbf{h}) - Q(\mathbf{w}) - \mathbf{g}^\top \mathbf{h}|}{\|\mathbf{h}\|_2} = 0.$$

The error in the linear approximation becomes negligible compared with the size of the input change.

One can prove that this vector is **unique** and equals the vector of partial derivatives, namely the **gradient**:

$$\mathbf{g} = \left(\frac{\partial Q}{\partial w_1}(\mathbf{w}), \dots, \frac{\partial Q}{\partial w_d}(\mathbf{w}) \right)^\top = \nabla Q(\mathbf{w})$$

Partial derivatives can exist even when the function is not differentiable.

The chain rule for a partial derivative

The ordinary chain rule still applies when we vary one argument and hold the other fixed.

$$R(a, b) = s^2, \quad s = a + 2b - 3$$

Vary a and hold b fixed.

$$\frac{\partial R}{\partial a} = \frac{d(s^2)}{ds} \frac{\partial s}{\partial a} = 2(a + 2b - 3) \cdot 1$$

Vary b and hold a fixed.

$$\frac{\partial R}{\partial b} = \frac{d(s^2)}{ds} \frac{\partial s}{\partial b} = 2(a + 2b - 3) \cdot 2$$

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- **Finding minima using differential calculus**

Local and global minima

We seek an input where the function has its smallest value.
The range of inputs used for comparison matters.

A local minimum compares nearby inputs.

$$f(x^*) \leq f(x) \quad \text{for all } x \text{ sufficiently close to } x^*$$

A global minimum compares all inputs in the domain.

$$f(x^*) \leq f(x) \quad \text{for every } x \text{ in the domain}$$

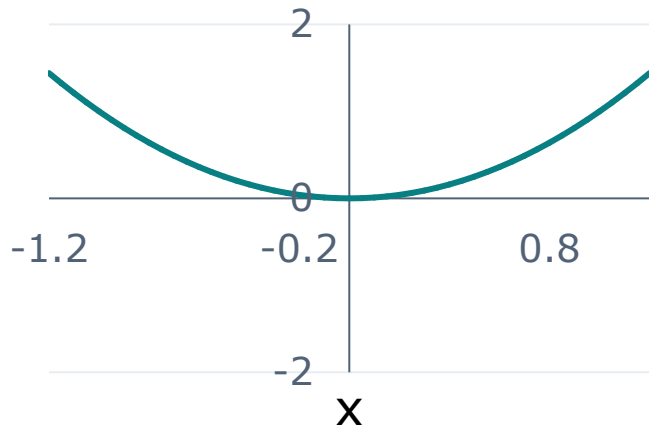
Every global minimum is also local. A local minimum need not be global.

Fermat's theorem in one variable

If f is differentiable at a local minimum x^* , then $f'(x^*) = 0$.

A point x such that $f'(x) = 0$ is called a **stationary point**

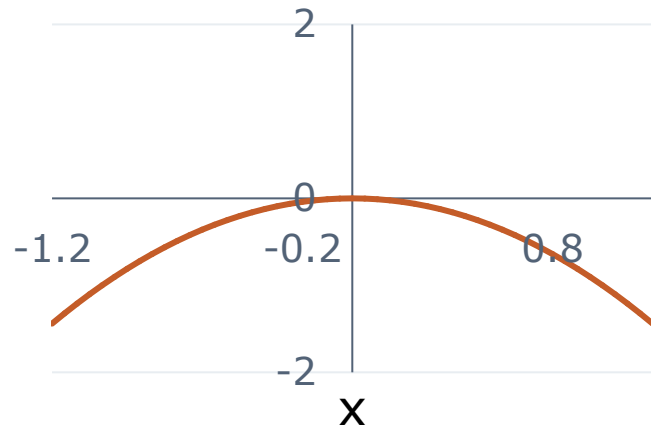
$$f(x) = x^2$$



$$f'(0) = 0$$

Minimum

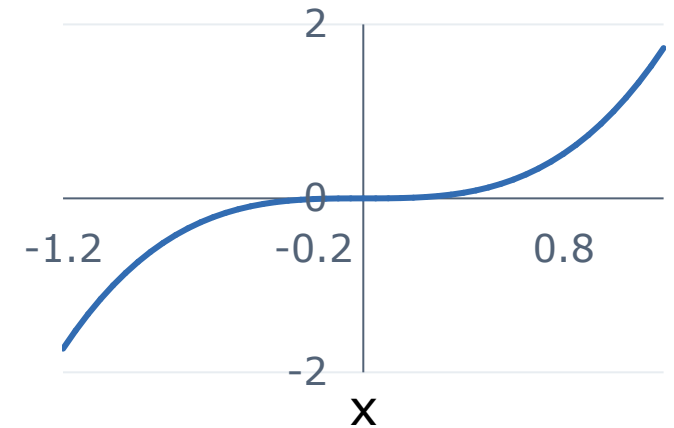
$$f(x) = -x^2$$



$$f'(0) = 0$$

Maximum

$$f(x) = x^3$$



$$f'(0) = 0$$

**Stationary
inflection point**

Being a **stationary point** is **necessary** for a local minimum, but it is **not sufficient**.

Finding and checking a minimum

The graph is an upward-opening parabola, so a global minimum exists.

$$F(a) = a^2 + 5a - 3 \quad (a \in \mathbb{R})$$

By Fermat's theorem, any minimiser must be a stationary point.

$$F'(a) = 2a + 5 = 0 \quad \Longleftrightarrow \quad a = -\frac{5}{2}$$

The minimum exists, and there is only one stationary point.
Therefore $a^* = -5/2$ is the global minimiser.

$$F\left(-\frac{5}{2}\right) = \frac{25}{4} - \frac{25}{2} - 3 = -\frac{37}{4}$$

Fermat's theorem in several variables

The same theorem holds in any dimension.
If H is differentiable at a local minimum $\mathbf{u}^*$, then

$$\nabla H(\mathbf{u}^*) = \mathbf{0}$$

For example, consider this function of two variables:

$$H(\mathbf{u}) = (u_1 - 1)^2 + (u_2 + 2)^2$$

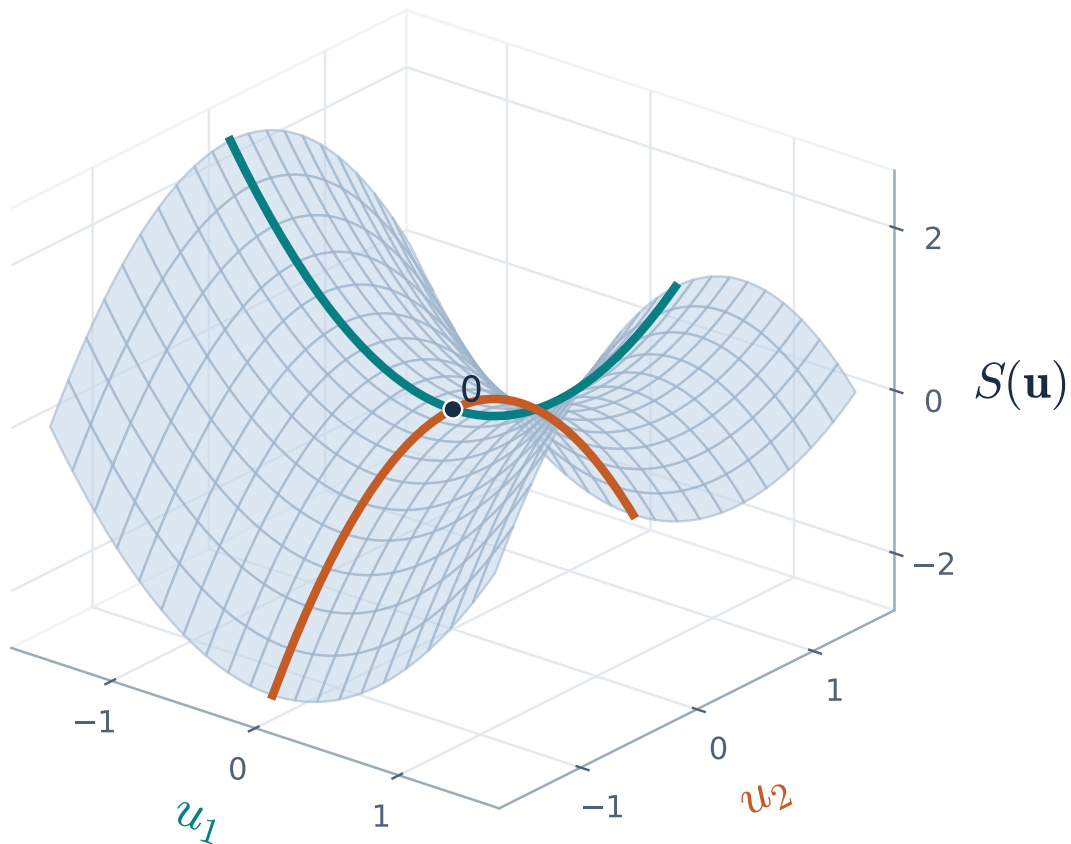
$$\nabla H(\mathbf{u}) = \begin{pmatrix} 2(u_1 - 1) \\ 2(u_2 + 2) \end{pmatrix} = \mathbf{0} \quad \implies \quad \mathbf{u}^* = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$H(\mathbf{u}) \geq 0, \quad H(\mathbf{u}^*) = 0$$

Here the stationary point is a global minimum: the sum of squares cannot be negative.

Saddle points

In one variable, stationary points may be minima, maxima or inflections.
In several variables, they may also be saddle points.



$$S(\mathbf{u}) = u_1^2 - u_2^2$$

$$\nabla S(\mathbf{u}) = \begin{pmatrix} 2u_1 \\ -2u_2 \end{pmatrix}$$

$$\nabla S(\mathbf{0}) = \mathbf{0}, \quad S(\mathbf{0}) = 0$$

$$S\begin{pmatrix} t \\ 0 \end{pmatrix} = t^2 > 0$$

$(t \neq 0)$

$$S\begin{pmatrix} 0 \\ t \end{pmatrix} = -t^2 < 0$$

Along one axis the origin is a minimum, along the other a maximum.

Lecture notes and exercises

The notes for today's lecture are available on the course GitHub page.

They explain the material in more detail and include exercises.

I will post the exercise solutions before the next lecture.

github.com/semtem0051/2026_27



Scan for course materials